

Experiment 5.1: Parallel forces

TOTAL: 15

Notes

Aim

In this experiment you will be able to test the principle of moments for yourself. In a number of simple equilibrium situations that you set up with the equipment provided you will be able to

- find the values for the clockwise and anticlockwise moments in each case you set up
- verify the principle of moments
- make predictions of where to hang masses on the apparatus in order to balance it
- calculate the mass of the metre rule that you use, by using the method of moments

Apparatus

- one metre rule with holes drilled at the 25 cm, 50 cm and 75 cm mark
- two sets of slotted 50 g masses
- 50 mm long bolt with a diameter of approximately 5 mm
- retort stand, boss head and clamp
- 0–10 N spring balance
- electronic pan balance
- wire or string for suspending masses from the metre rule
- two bulldog clips or similar could also be used to support masses

Pre-lab

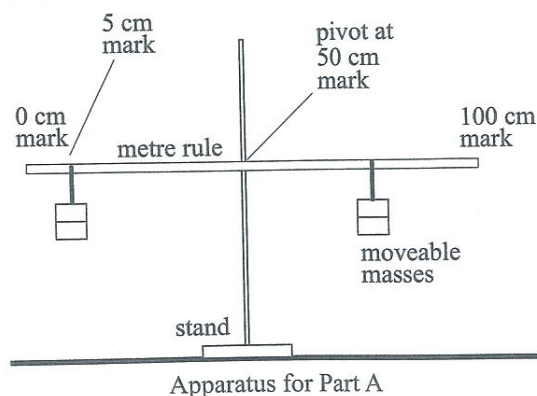
Part A: Balancing a constant moment

- In this part of the experiment you will balance a constant moment produced by a constant weight that is at a fixed distance from the pivot. To do this you will need to use different masses at varying distances from the pivot.
- Set up the equipment as shown by first placing the bolt through the rule, then clamping the bolt with the boss-head. Make sure the rule balances horizontally before you add any weights.
- Adjust the height of the pivot so that when the rule is balanced the weights are a few centimetres above the bench top.

Note: If the metre rule represented is balanced in the horizontal position, the 100 g mass provides an anticlockwise moment about the bolt. The moveable masses provide an equal magnitude moment in a clockwise direction on the other side of the pivot. When the anticlockwise moment cancels the clockwise moment about the bolt, the rule and the masses will be in equilibrium.

Fixed Mass Data	
Mass in grams (g)	100 g
Mass in kilograms (kg)	
Distance from pivot (cm)	45.0 cm
Distance from pivot (m)	
Weight in newtons (N)	
a c moment about pivot (Nm)	

Table 1



Experiment 5.1: Parallel forces

5

Lab notes

- Hang a 100 g mass between the 50 cm mark and the 100 cm mark in a position so that the metre rule is balanced in the horizontal position.
- Use the scale on the metre rule to measure the distance from the pivot to the point where you hung the second 100 g mass and record this value in table 2 as shown below, in the column labelled 'Distance to weight (cm)'.
- Add successive 50 g masses and repeat steps 2 and 3 for all mass values shown in the first column of table 2.

Post-lab discussion

Mass used (g)	Mass used (kg)	Weight used (N)	Distance to weight (cm)	Distance to weight (m)	weight $\times$ distance (N m)
100					
150					
200					
250					
300					
350					
400					
450					

Table 2

Fill in the remaining blank spaces in table 1 and table 2.

- Comment on any regularity in the values you obtain in the last column of table 2.
- Compare the average value in table 2 to the value of the anticlockwise moment in table 1.
- Write a statement about moments that summarises your findings in this part of the experiment.
- Why is the mass of the rule ignored in your experiment?
- Comment on how the length of the lever arm for the clockwise moment changes as you add more weights at a fixed distance to the left of the pivot. Explain your reasoning.

Increasing r - $F \times r$ needs to be larger

Table 2
sig fig
units

Does $cm = 4cm$?

At the pivot - no turning effect

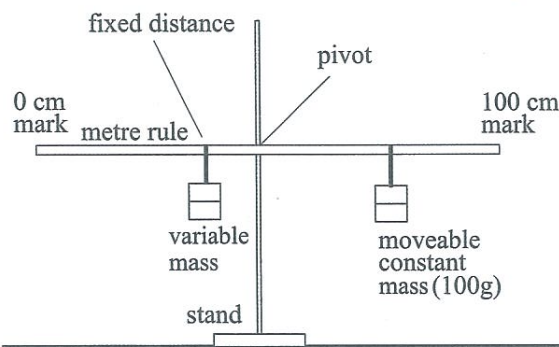
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Pre-lab

Part B: Balancing a variable moment

In this part of the experiment you will balance a changing moment produced by a changing weight that is at a fixed distance from the pivot. To do this you will need to shift a constant mass to varying positions from the pivot.

Set up the equipment as in Part A with a 100 g mass at the 40 cm mark on the rule. This will be at the fixed distance of 10 cm from the pivot for this part of the experiment. See the diagram (right).



Apparatus for Part B

Experiment 5.1: Parallel forces

Notes

Lab notes

- Position a 100 g mass at the 40 cm mark and balance the rule by hanging a 100 g mass on the other side of the pivot. Record the distance to the pivot of the second mass in table 3.

Values for Constant Measurement	
Mass of movable constant weight (g)	100 g
Weight of movable constant (N)	
Distance from pivot of variable mass (cm)	10.0 cm
Distance from pivot of variable mass (m)	

Table 3

- Add a further 50 g mass to the first one at the 40 cm mark and balance the rule by moving the 100 g constant mass. Record the new balance position.
- Repeat steps 2 and 3 for each of the mass values shown in the first column of table 4.

Mass used at fixed distance (g)	Weight used at fixed distance (N)	Distance to pivot for constant weight (cm)	Distance to pivot for constant weight (m)	Moment for constant weight (N m)	Moment for variable weights (N m)	Difference between ACM and CM (N m)
100						
150						
500						
250						
300						
350						
400						
450						

Table 4

Post-lab discussion

- Fill in all the remaining blank spaces in table 3 and table 4.
- Comment on any regularity in the values you obtained in the last column of table 4.
- Write a statement about moments that summarises your findings in this part of the experiment.
- Why can you ignore the mass of the rule in your experiment?
- Comment on how the length of the lever arm for the clockwise moment changes as you add more weights at a fixed distance to the left of the pivot. Explain your reasoning.

Table 2
sig fig
units

Does $CM = ACM$?
At pivot
Gets larger

3

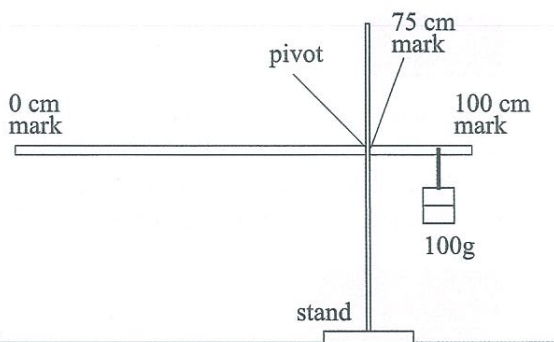
Experiment 5.1: Parallel forces

5

Pre-lab discussion

Part C: Weighing a metre rule

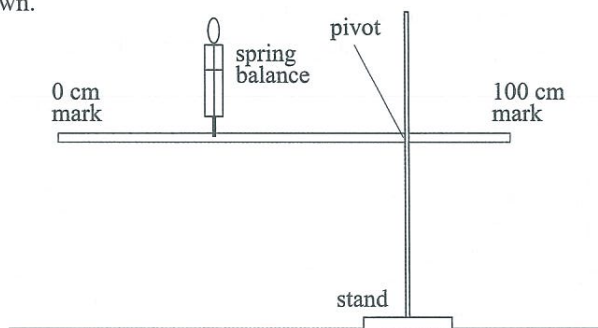
- In this part of the experiment you will calculate the weight of the metre rule by using the principle of moments when the rule is balanced. To do this you will need to change the pivot point of the rule.
- Set up the equipment as shown by placing the bolt through the rule at the 75 cm mark.



Apparatus for Part C

Lab notes

- Hang a 100 g mass at a point on the rule so that the rule comes to rest in a horizontal position. Record the distance to the pivot of the 100 g mass.
- Mark on the diagram the position of the ruler's centre of mass and all the forces acting on the rule.
- Use your knowledge of moments to calculate the weight of the rule. Record this value.
- Remove the 100 g mass and use a spring balance to support the rule horizontally from its 30 cm mark as shown.



Using a spring balance

- Mark on the diagram all the forces acting on the rule.
- Use the reading from the spring balance to calculate the weight of the rule. Record this value.
- Dismantle your apparatus and weigh your rule on an electronic balance. Record this value.

Post-lab discussion

- Compare the two values you obtained for the weight of the rule and comment on any differences.
- Would the reading on the spring balance change if the rule was not supported horizontally? Explain your answer. **Yes - smaller. + decreases.**
- If the pivot in "Using a spring balance" was replaced by another spring balance, what reading would it show? Use the measured weight of the rule you found from the electronic balance to calculate your answer. **Mass ruler + mass brass**

Calculation of mass

(2)

Same?

(3)